

1h 39m
left

ALL



1

1. Valid BST Permutations

A binary tree is a data structure characterized by the following properties:

- It can be an empty tree where root = null.
- It can consist of a root node that contains a value and two sub-trees labeled "left" and "right".
- Each sub-tree also follows the definition of a binary tree.

2

A binary tree is a binary search tree (BST) if all the non-empty nodes follow these two rules:

3

- If a node has a left sub-tree, then all the values in its left sub-tree are smaller than its value.
- If a node has a right sub-tree, then all the values in its right sub-tree are greater than its value.

4

5

Given an integer, determine the number of valid BSTs that can be created by nodes numbered from 1 to that integer. Please see the samples below for diagrams based on 1, 2 or 3 nodes.

6

7

Function Description

Complete the function *numBST* in the editor below.

8

numBST has the following parameter(s):

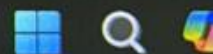
Language C++20

```

1 > #include
9
10 /*
11  * Comple
12  *
13  * The fu
14  * The fu
15  */
16
17 vector<in
18
19 }
20
21 > int main

```

Test Results



4. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

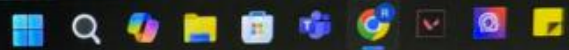
☒ This memory address does not map to actual memory cell that corresponds to address number.

☐ This memory is the same as physical memory.

☐ This memory cannot be accessed by applications at run time.

☐ This memory address is always present even after a system is restarted.

Clear Selection



MSI



ALL

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2. Which interrupt is defined for system calls in Linux?

Which interrupt is defined for system calls in Linux?

Pick **ONE** option

☐ 0x10

☐ 0x60

☒ 0x80

☐ 0x40

Clear Selection



MSI



5. What is the postfix expression for the corresponding infix expression? $a+b*c+(d*e)$

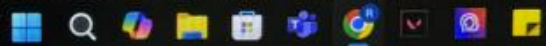
What is the postfix expression for the corresponding infix expression? $a+b*c+(d*e)$

Pick **ONE** option

- ☒ $abc*+de*+$
- ☐ $abc*+de*+$
- ☐ $a+bc*de*+$
- ☐ $abc*+(de)*+$

Clear Selection

6. #include <stdio.h> #define MAX(a,b) ((a)>(b)?(a):(b)) int main() { int x = 5, y = 10; printf("



MSI



3. Operating system maintains the information about a process in which of the following?

Operating system maintains the information about a process in which of the following?

Pick **ONE** option

☐ Heap

☒ Process Control Block

☐ Stack

☐ TLB Cache

Clear Selection



MSI

1h 39m
left

ALL



1

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8

BSTs that can be created by nodes numbered from 1 to that integer. Please see the samples below for diagrams based on 1, 2 or 3 nodes.

Function Description

Complete the function *numBST* in the editor below.

numBST has the following parameter(s):

int nodeValues[n]: the number of node values to analyze

Returns:

int[n]: the number of different binary search trees that can be created for each test case modulo 100000007 ($10^8 + 7$).

Constraints

$1 \leq n \leq 1000$

$1 \leq \text{nodeValues}[i] \leq 1000$

► Input Format for Custom Testing

▼ Sample Case 0

Sample Input

STDIN	Function
5	→ <i>nodeValues</i> size $n = 5$
1	→ <i>nodeValues</i> = [1, 2, 3, 4, 100]

Language C++20

```
1 > #include <
9
10 /*
11 * Complet
12 *
13 * The fur
14 * The fur
15 */
16
17 vector<int
18
19 }
20
21 > int main()
```

Test Results

8. Consider a database table "orders" with columns: order_id (Primary Key), customer_id, order_date, and total_amount. Which of the following SQL statements would correctly create a partitioned table based on the order_date column, partitioned by range?

Consider a database table "orders" with columns: order_id (Primary Key), customer_id, order_date, and total_amount. Which of the following SQL statements would correctly create a partitioned table based on the order_date column, partitioned by range?

Pick **ONE** option

- ☐ create table orders partition by range columns (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);
- ☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2) primary key (order_id);
- ☒ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);
- ☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2) partition by range (order_id);

Clear Selection



7. Consider the below java method. What's the output of this method? `public static void function_1(int [] data, int low, int high) { if (low < high) { int temp = data[low]; data[low] = data[high]; data[high] = temp; function_1(data, low +1, high -1); } }`

ALL

1

Consider the below java method. What's the output of this method?
`public static void function_1(int [] data, int low, int high) {
 if (low < high) {
 int temp = data[low];
 data[low] = data[high];
 data[high] = temp;
 function_1(data, low +1, high -1);
 }
}`

4

5

6

Pick ONE option

7

☐ The sum of all the elements of the array

8

☐ The product of all the elements in the array.

9

☐ The last element of the array.

10

☒ Reverses the contents of the array.

Clear Selection

6. `#include <stdio.h> #define MAX(a,b) ((a) > (b) ? (a) : (b)) int main() { int x = 5, y = 10; printf(" %d %d %d\n", MAX(x++, y++), x, y); return 0; }`

ALL

1

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4

5

6

7

8

9

```
#include <stdio.h>
#define MAX(a,b) ((a) > (b) ? (a) : (b))

int main() {
    int x = 5, y = 10;
    printf(" %d %d %d\n", MAX(x++, y++), x, y);
    return 0;
}
```

Pick ONE option

☐ 11 6 12

☐ 12 6 12

☐ 11 7 12

☒ 12 7 12

Clear Selection



ALL

9. Which of the following is a key-value store NoSQL database system?

Which of the following is a key-value store NoSQL database system?

Pick **ONE** option

6

☐ HBase

7

☐ MongoDB

8

☒ Redis

9

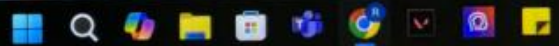
☐ Accumulo

10

Clear Selection

11

12



MSI



ALL

10. What is the latest long-term release of Oracle Database?



What is the latest long-term release of Oracle Database?

Pick **ONE** option

7

☐ 23ai

8

☐ 22ai

9

☒ 24ai

10

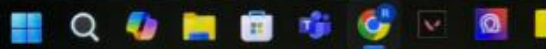
☐ 19ai

11

Clear Selection

12

13



MSI

ALL

12. What does CAP theorem describe with respect to System Design ?

What does CAP theorem describe with respect to System Design ?

Pick **ONE** option

- ☐ The relationship between CPU and memory usage
- ☒ The trade-offs between consistency, availability, and partition tolerance
- ☐ The process of data encryption
- ☐ The principles of object-oriented programming

Clear Selection





ALL



11. Which of the following is not a property of Relational Database?

Which of the following is not a property of Relational Database?

Pick **ONE** option

☐ Atomicity

☐ Consistency

☐ Isolation

☒ Dependency

Clear Selection



MSI

ALL

13. Which of the following design criteria are not valid/suitable for Non-Relational Databases?

Which of the following design criteria are not valid/suitable for Non-Relational Databases?

Pick ONE option

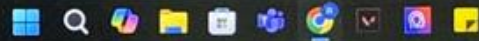
☐ Your application required super-low latency

☐ Your data are unstructured without any relation between them

☐ You only need to serialize & deserialize data (JSON, XML, YAML)

☒ You need to store a very small amount of data

Clear Selection



MSI

16. What does idempotency mean in REST API design?

What does idempotency mean in REST API design?

Pick **ONE** option

- ☐ The ability of an API to handle multiple requests in parallel
- ☒ The property that a method can be called multiple times without different outcomes
- ☐ The capability of an API to update data
- ☐ The feature of an API that allows it to delete resources

Clear Selection



14. What is the use of /etc/hosts file in Linux ?

What is the use of /etc/hosts file in Linux ?

Pick **ONE** option

- ☒ It contains the list of all users.
- ☐ It contains a list of all files in the system
- ☐ It contains the time and date information.
- ☐ It contains the FQDN to IP address mappings.

Clear Selection





ALL

18. In a ring topology network, if one device fails, how does it affect the network?

In a ring topology network, if one device fails, how does it affect the network?

Pick **ONE** option

15

☐ Only the failed device is affected

16

☐ All devices in the network are affected

17

☒ Only the devices between which the failed device exists are affected

18

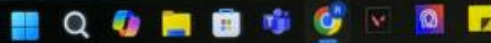
☐ The network continues to function normally

19

Clear Selection

20

21



MSI

HackerRank Question - Wh...
hackerrank.com/test/9b18hg986s/questions/966nc2b4fp
Dean Academic | M... WhatsApp RESOURCE HUB - G... Data Analyst Roadm... Excel Practice Onlin... SQL Tutorial Learn Python - Free... (2) Data Analyst Por...
ORACLE FY26 - Campus Hiring -PDO_ MNIT Allahabad(FullTime)_8 Aug Answered: 2 / 36 01 hour 36 m

ALL 17. Which is the HTTP request method used to retrieve the metadata associated with the resource's state ?

Which is the HTTP request method used to retrieve the metadata associated with the resource's state ?

Pick **ONE** option

14 ☐ GET

15 ☐ PUT

16 ☒ HEAD

17 ☐ POST

18 Clear Selection

19

20



MSI





ALL

15. Which Clause restricts the groups of rows returned to those groups for the specified condition id True



Which Clause restricts the groups of rows returned to those groups for the specified condition id True

12

Pick **ONE** option

13

☐ Where clause

14

☒ Having Clause

15

☐ Distinct

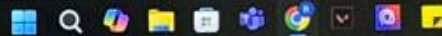
16

☐ Exists

17

Clear Selection

18



MSI

20. In what way the Symmetry Sparse Matrix can be stored efficiently?

In what way the Symmetry Sparse Matrix can be stored efficiently?

Pick ONE option

- ☐ Heap
- ☒ Binary tree
- ☐ Hash table
- ☐ Adjacency List

Clear Selection



MSI

ALL

19. What is the purpose of ARP (Address Resolution Protocol) in computer networking?

What is the purpose of ARP (Address Resolution Protocol) in computer networking?

Pick **ONE** option

16

☒ To convert IP addresses to MAC addresses

17

☐ To convert domain names to IP addresses

18

☐ To establish a secure connection between client and server

19

☐ To route packets between networks

20

Clear Selection

21

22



MSI



ALL



21. Which of the following is false?

Which of the following is false?

Pick **ONE** option

18

19

20

21

22

23

24

☐ 2-3 tree requires less storage than the BST

☐ lookup in 2-3 tree is more efficient than in BST

☐ 2-3 tree is shallower than BST

☐ 2-3 tree is a balanced tree

Clear Selection

Continue



MSI

22. Chef's Puzzle 1

Out of the nine ingredients, a chef wants to use exactly five. Given the following statements, how many of the 5 ingredients are known?

The ingredients are apricots, bacon, cake, donuts, eggs, figs, grapes, hazelnuts, and ice cream.

1. If bacon is not used, then grapes, hazelnuts, and ice cream are not used.
2. If apricots are used then bacon and cake are used.
3. If donuts are not used, apricots are not used.
4. Apricots and eggs are used.

Pick **ONE** option

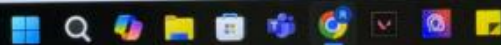
☐ 2

☐ 3

☐ 4

☐ 5

Clear Selection



MSI



ALL

24. Painted Cube

A cube is painted green on all surfaces and is cut into 1000 identical cubes. How many of the smaller cubes have three faces painted?

Pick **ONE** option

21

☒ 8

22

☐ 10

23

☐ 100

24

☐ 64

25

[Clear Selection](#)

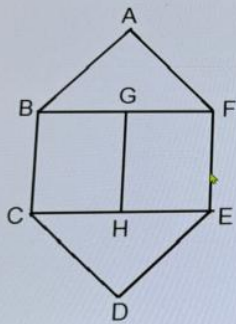
26

27



23. Paths in a Graph 2

There is a network of eight cities (represented by vertices) connected by roads. A traveler must drive on each road segment once but can visit a city multiple times if necessary. Furthermore, the starting and ending points do not have to match. New roads are constructed between C and G, and between H and F. If the journey begins at B, where will it end?



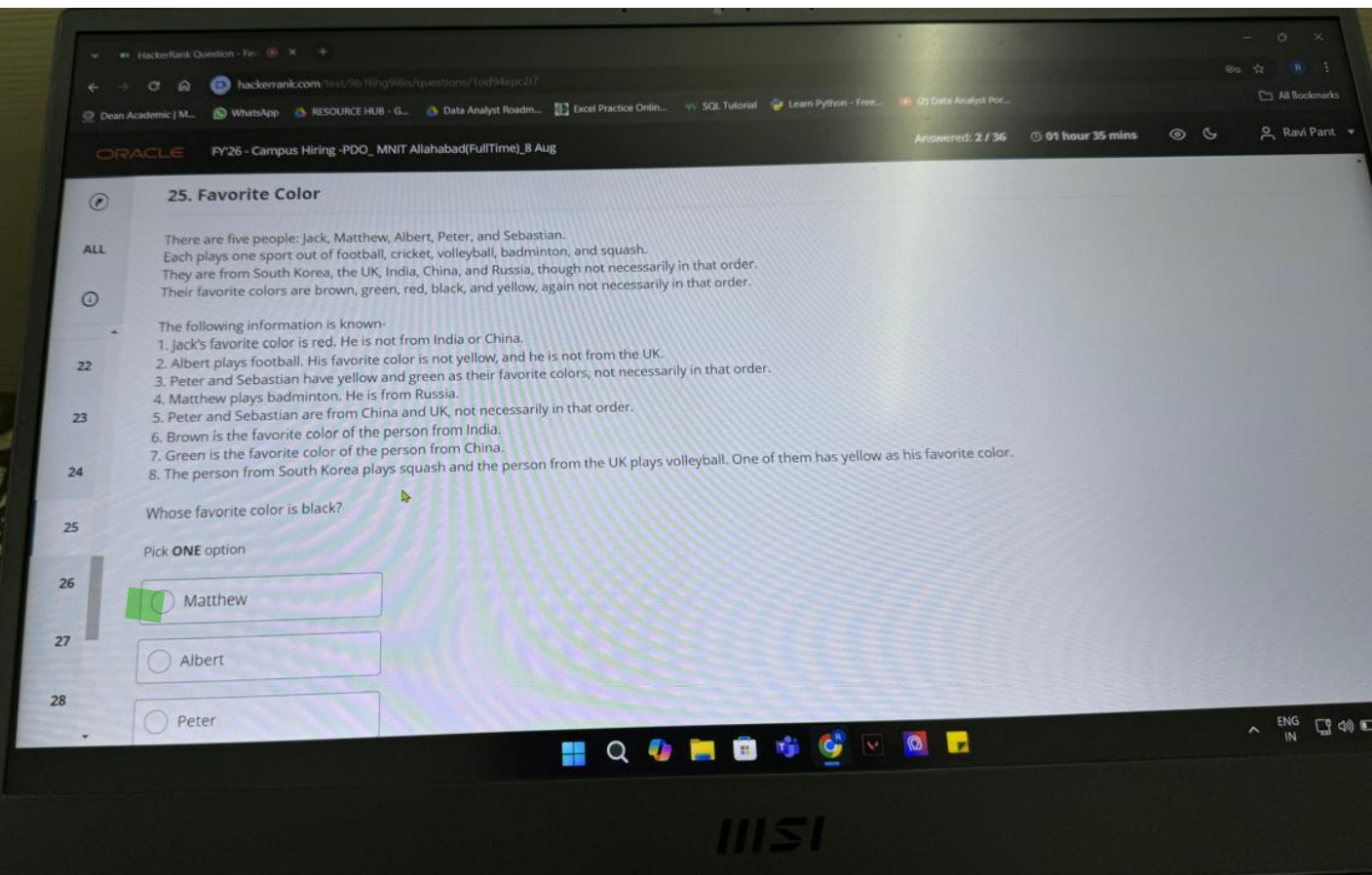
Pick ONE option

27. What Comes Next? 4

Predict the figure in (?) place.

Answer Fig:

(1) (2) (3) (4)



26. Standing Order 2

ALL

There are five people - Peter, Tyson, Richard, Steve and Quinn - standing in a row. Given the following statements, who is in the middle?

1. Peter is next to Quinn and Steve is next to Richard.

1. Peter is next to Quinn and Steve is next to Richard.
2. Steve is not next to Tyson.
3. Tyson is on leftmost position.
4. Richard is on the second position from the right.
5. Peter stands somewhere to the right of both Quinn and Tyson.
6. Peter and Richard are next to each other.

Pick **ONE** option

24

25

26

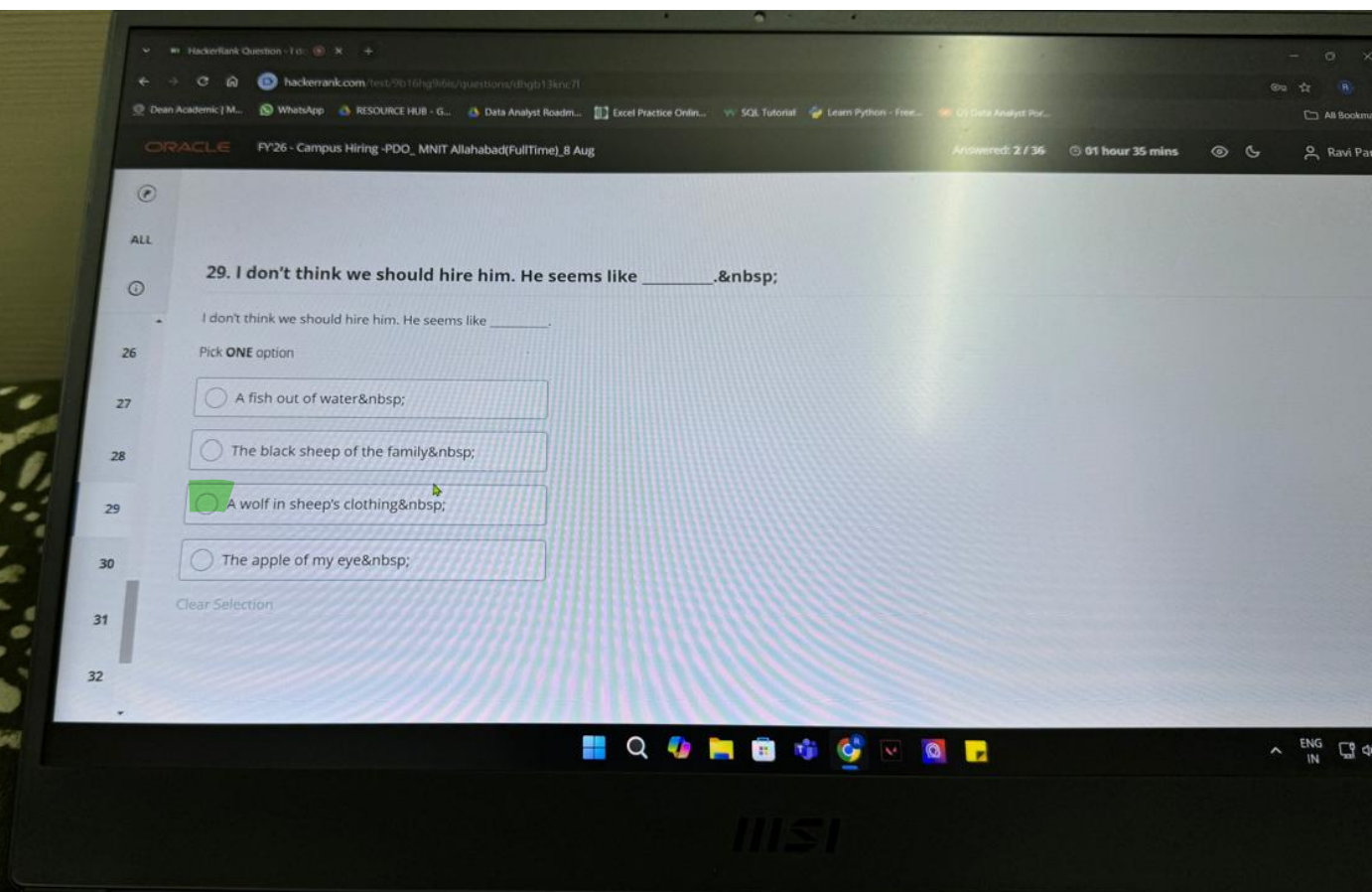
7

 Peter

☐ Richard

☐ Steve

☐ Quin



28. In the following question, choose the correct option which can be substituted for the given sentence. A self-governing country or region

In the following question, choose the correct option which can be substituted for the given sentence.
A self-governing country or region

Pick ONE option

- 25 ☒ Autonomy
- 26 ☐ Autocracy
- 27 ☐ Anarchy
- 28 ☐ Ethnology

Clear Selection

31. Reading Comprehension - The Second Industrial Revolution in America V

Read the passage and answer the questions that follow.

While the first industrial revolution, which spanned the years from roughly 1760 to about 1820-1840, laid the groundwork, the second industrial revolution brought about changes truly desired by both the masses and high-profile citizens. The second industrial revolution, as defined by historians, took place between 1870 and 1914. At the time, there was an abrupt rise in urbanization and the development of additional land. This created the need for technological breakthroughs to make living simpler. The steel industry boomed while iron became almost obsolete. Steel's lower cost and higher strength-to-weight ratio made it simpler and more practical for building projects, industrial machinery, railroads, ships, and many other things. As a result, building bigger structures, creating more mechanical commodities, and traveling and transporting goods farther distances became common.

The development of electricity into wide use was even more significant. Before the arrival of public electricity, candles and gas lamps were used to light houses and factories. Electric lighting altered how people worked and lived, as working hours were no longer restricted to daytime. Beyond a source of illumination, public electricity spawned numerous other inventions. Alexander Graham Bell invented the telephone in 1876. This inspired Guglielmo Marconi to invent the practical radio wave-based wireless telegraph system that could send signals over the Atlantic Ocean in 1901. Thomas Edison perfected the light bulb in 1879, which was reported to last 13 and a half hours in the wake of several unsatisfactory attempts by numerous inventors.

The first electric railroad debuted in Berlin, Germany, in 1879. Likewise, electric streetcars began to replace horse-drawn carriages in major European cities as early as 1880. The crown of the second industrial revolution was the internal combustion engine. It was first presented in 1878. The gas and the air-driven vehicle were unsuitable for the general public due to safety issues. With the advent of more stable liquid fuels such as petroleum, initial problems were quickly overcome. Automobiles and airplanes were developed as a result of this engine. None of the following innovations would have been conceivable without it, implying that the modern means of transportation would be only a dream.

The German inventor Karl Benz made the first self-propelled car in 1886. It was made available to the public in 1888. In the same year, Gottlieb Daimler developed a lighter-weight engine, and in 1896, Henry Ford fully commercialized the use of automobiles. Years later, the Wright brothers invented the first practical aircraft, overcoming the difficult technical issues that had impeded mechanical flight for centuries. Before their invention, steerable or dirigible hydrogen-powered balloons were the only method of air travel available to man.

Since then, the world has experienced several industrial revolutions that have paved the way for the development of electronics, telecommunications, and computers and opened the door for space exploration, research, and biotechnology. Still, the second revolution is still remembered as the era of iconic inventions that spawned technology use today.

The word "spawned" in the passage is farthest in meaning from

Pick ONE option



	46.18		47.34		49.32
	Aug-12		Sep-12		Oct-12

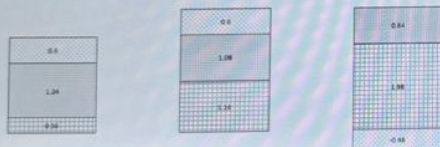
Customers in Millions Using Each Phone

	Hington	Euphore
Aug-12	46.18	47.28
Sep-12	47.34	48.91
Oct-12	49.32	49.27

The following chart shows the addition of customers (in millions) of Hington and Euphore in each of the three months. There are two versions of the Euphore brand phones, Version-1, and Version-2.

Addition of customers (in millions)

☐ Hington ☐ Version-1 ☐ Version-2

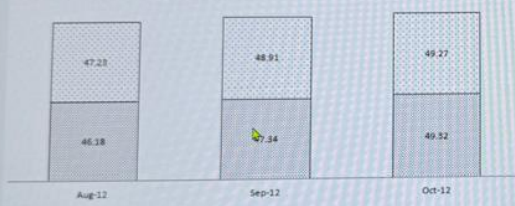


35. Interpret Charts

The following graph shows the number of people (in millions) using Hington and Euphore brand phones in France from August to October 2012. Only these two brands of phones exist in the market.

Number of customers (in millions) using each brand

□ Hington □ Euphore



Customers in Millions Using Each Phone

	Hington	Euphore
Aug-12	46.18	47.28
Sep-12	47.34	48.91
Oct-12	49.52	49.27

ALL

29

30

31

32

33

34

35

36

Aug-12 Sep-12 Oct-12

Addition of Customers in millions

	Hington	Version-1	Version-2
Aug-12	0.36	1.24	0.6
Sep-12	1.16	1.08	0.8
Oct-12	1.98	0.84	0.48

What was the total number of people using either of the two brands in July 2012?

Pick ONE option

- ☐ 91.21 million
- ☐ 6.87 million
- ☐ 98.23 million
- ☐ None of these



Clear Selection

ALL



29

34. Integer Division

30

What is the remainder of 1992 divided by 92?

31

Pick **ONE** option

32

☐ 0

33

☐ 1

34

☐ 40

35

☒ None of the above

60

36

Clear Selection

1. Binary Manipulation

Given an integer, store its binary representation *bin* as a string or an array with digits indexed from 0 to $\text{length}(\text{bin}) - 1$. Reduce its binary representation to zero by using the following operations:

1. Change the i^{th} binary digit only if $(i+1)^{\text{th}}$ binary digit is 1 and all other binary digits from $(i+2)$ to the end are zeros.
2. Change the rightmost digit without restriction.

What is the minimum number of operations required to complete this task?

Example

$n = 4$

The binary representation of 4 is 100 so for the example, $\text{bin} = [1, 0, 0]$.

100 $\rightarrow$ 101 $\rightarrow$ 111 $\rightarrow$ 110 $\rightarrow$ 010 $\rightarrow$ 011 $\rightarrow$ 001 $\rightarrow$ 000

bin, from left to right, has indices 0 to 3. Stepping through the operations:

- Use rule 2 to change $\text{bin}[2]$ to 1. 100 $\rightarrow$ 101
- Use rule 1 to change $\text{bin}[1]$ to 1. 101 $\rightarrow$ 111

Language C++20 Environment

```
1 > #include <bits/stdc++.h>...
9
10 /*
11  * Complete the 'minOperations' function
12  *
13  * The function is expected to return an integer.
14  * The function accepts LONG_INTEGER n as parameter.
15  */
16
17 long minOperations(long n) {
18
19 }
20
21 > int main()...
```

Test Results

Custom Input



Search



Ideapad GAMING

1h 39m left



ALL



Function Description

Complete the function `minOperations` in the editor below.

`minOperations` has the following parameter:

long `n`: the starting value in base 10

Returns:

int: the minimum number of operations required to convert `n` to 0

Constraints

- $1 \leq n \leq 10^{15}$

Input Format For Custom Testing

Sample Case 0

Sample Input For Custom Testing

STDIN	Function
13	n = 13

Sample Output

9

Explanation

The binary representation of 13 is 1101. This is the sequence of the 9 operations that change 13 to 0 according to the constraints:

Language C++20

Environment

```
1 > #include <bits/stdc++.h> ...
9
10 /*
11  * Complete the 'minOperations' function
12  *
13  * The function is expected to return an integer.
14  * The function accepts LONG_INTEGER n as parameter.
15  */
16
17 long minOperations(long n) {
18
19 }
20
21 > int main() ...
```

Test Results

Custom Input



Search



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32. Reading Comprehension - The End of the French Monarchy V

Read the passage and answer the questions that follow.

As the 18th century came to a close, France and Britain were in a drawn-out war to advance their empires. Due mainly to the cost of war, France was on the verge of bankruptcy. Meanwhile, French King Louis XVI and his queen, Mary Antoinette, lived an extravagant lifestyle. Despite the fact that he ruled France during the height of the Enlightenment Era, Louis XVI was unable to establish himself as an enlightened absolutist. His efforts to allow citizens more opportunities along with some proposed societal ideals led to increased bureaucracies and even more expenditures. The royal treasury was almost empty, necessitating additional taxes.

At the same time, there were years of poor harvests due to drought. Spiking bread prices led to unrest among the non-aristocrat classes. This, coupled with increased taxes, caused resentment among the lower classes. Even some in the noble classes became disenchanted with the monarchy as their own access to a lavish lifestyle was curtailed. As the war continued, citizens became agitated and began to speak out against the weakened monarchy. The excesses of the monarchy showed citizens that any spoils of war were not likely to be shared with them. Quickly, their unrest changed from passive to active, and common people began plans to depose the monarchy.

The nation was overtaken by a wave of mass panic and revolutionary passion. Peasants robbed and set fire to the homes of tax collectors, landlords, and members of the aristocratic class. Out of concern for his family's safety and disgust with the ongoing revolution in the country, in June 1792, the king fled Paris for the Austrian frontier. During the journey, he was recognized, captured, and returned to Paris. An irate public demanded his imprisonment in August 1792, and the monarchy was overthrown in September.

Soon after, evidence of the king's counter-revolutionary plots with other countries was discovered, and he was charged with treason. He was found guilty in January 1793, and given the death penalty. He stubbornly approached the guillotine on January 21 and was put to death. Nine months later, queen Marie-Antoinette was executed for the same reason. Ultimately, the monarchy's end resulted from the royals' penchant for abusing power. Decades of waste and obscene indulgences ended in a brutal, extreme manner by a people no longer willing to be yoked. Perhaps the monarchy would still exist if they had paid more attention to the concerns of the populace.

According to the passage, which of the following statements is incorrect?

Pick ONE option

☒ All the nobles were happy with the way King Louis XVI governed the country.

☐ The underclass plundered and pillaged the homes of nobles during the height of the unrest.

1h 39m left

Example

$n = 4$

The binary representation of 4 is 100 so for the example, $bin = [1, 0, 0]$.

ALL

100 → 101 → 111 → 110 → 010 → 011 → 001 → 000

i

bin , from left to right, has indices 0 to 3. Stepping through the operations:

1

- Use rule 2 to change $bin[2]$ to 1. 100 → 101

2

- Use rule 1 to change $bin[1]$ to 1. 101 → 111

3

- Use rule 2 to change $bin[2]$ to 0. 111 → 110

4

- Use rule 1 to change $bin[0]$ to 0. 110 → 010

- Use rule 2 to change $bin[2]$ to 1. 010 → 011

- Use rule 1 to change $bin[1]$ to 0. 011 → 001

- Use rule 2 to change $bin[2]$ to 0. 001 → 000

5

For this number, 7 operations are required to convert it to 0.

6

Note: In the binary representation of a number, the binary digit's positions are numbered as 0 to $x-1$ from left to right, where x is the number of digits in the binary representation of the number.

7

Function Description

8

Complete the function `minOperations` in the editor below.

Language C++20

Environment

```
1 > #include <bits/stdc++.h>...
9
10 /*
11  * Complete the 'minOperations' function.
12  *
13  * The function is expected to return a long integer.
14  * The function accepts LONG_INTEGER n as parameter.
15  */
16
17 long minOperations(long n) {
18
19 }
20
21 > int main()...
```

Test Results

Custom Input



Search



IdeaPad GAMING





ALL



29

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36. How Many Like Cricket?

In a group of students, 60% like football, 40% like cricket. What percentage of students like only cricket if 12% of the students like neither football nor cricket?

Pick **ONE** option

☒ 28

☐ 14

☐ 56

☐ 42

Clear Selection

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Answered: 2 / 36 01 hour 33 mins Ravi Pant

ALL

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33. Logic - Sports Enthusiasts

In a company,100 employees are randomly selected. 50 like watching football 40 like watching hockey and the rest like both. Find out the number of employees who like to watch at least one of the games.

Pick ONE option

☐ 40

☐ 90

☒ 80

☐ 20

Clear Selection

Windows Taskbar

19:09 08-08-2024